Prodigy Anesthesia Quick Review

Substance Abuse

Define drug tolerance.	Drug tolerance is a condition in which the body's tissues become so adjusted to the presence of a drug that increased doses of the drug are necessary to produce the desired effect. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 542.*
What is physical dependence to a drug?	Physical dependence to a drug is a state in which the drug must be present in the body in order to prevent symptoms of withdrawal so that normal physiologic activity can occur. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 542.*
What medical problems are associated with the chronic abuse of opioids?	Malnutrition, aspiration pneumonitis, cellulitis, hepatitis, endocarditis, tetanus, superficial skin abscesses, transverse myelitis, acquired immunodeficiency syndrome, septic thrombophlebitis, systemic septic emboli *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 546-548.*
What is the number one cause of unconsciousness seen in individuals admitted to the emergency room?	Drug overdose *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 543.*
What are some contraindications to gastric lavage?	Gastric lavage is not advised in patients in which laryngeal reflexes are not intact, or when the substance ingested is corrosive or a hydrocarbon. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 543.*
When are signs and symptoms of early alcohol withdrawal syndrome typically seen?	6 to 8 hours after a drastic reduction in the blood alcohol concentration. These signs and symptoms are most pronounced at 24-36 hours. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 544.*
What are the two major risk factors associated with alcoholism.	A family history of alcoholism, male gender *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 544.*
What are the signs and symptoms of early withdrawal syndrome?	Hyperactivity of the autonomic nervous system (cardiac dysrhythmias, high blood pressure, tachycardia), generalized tremors, perceptual disturbances such as hallucinations and nightmares, mild confusion with associated agitation, insomnia, nausea, and vomiting. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 544.*

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What is delirium tremens?	A medical emergency that is life-threatening and is characterized by combativeness, tachycardia, grand mal seizures, hallucinations, and hypertension or hypotension. Delirium tremens is seen 2-4 days once the ingestion of alcohol has stopped. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 544.*
What is the mortality rate of delirium tremens? What is the typical cause of death?	The mortality rate of delium tremens is 10% and is primarily caused from seizures, hypotension, or cardiac dysrhythmias. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 544.*
How does cocaine stimulate the sympathetic nervous system?	Cocaine blocks the presynaptic uptake of dopamine and norepinephrine, resulting in increased postsynaptic concentrations of the neurotransmitters. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 545.*
What are the characteristics of fetal alcohol syndrome?	Mental retardation, growth retardation, and craniofacial dysmorphology *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 545.*
How does an overdose of aspirin exert its pathologic effects within the body?	Once in the body, aspirin is converted to its active metabolite, salicylic acid. Salicylates are metabolic poisons at toxic levels, affecting many organ systems by uncoupling oxidative phosphorylation and interfering with the Krebs cycle. Ketoacids and lactic acids accumulate within the body as a result of the oxidative phosphorylation uncoupling. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 552-553.*
Describe the treatment of cocaine overdose.	The treatment for patients with an overdose to cocaine should include the management of myocardial ischemia through the administration of nitroglycerin. If cardiac vasospasm is present, treatment with an alpha adrenergic blocker is suggested, as beta adrenergic blockade can accentuate this condition. Seizures should be managed through the administration of IV benzodiazepines, and if hyperthermia is present, active cooling measures are indicated. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 545.*
What is the primary manifestation of an overdose associated with barbiturates?	Central nervous system depression *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 548.*
What is the most serious complication associated with the withdrawal of barbiturates?	Grand mal seizures *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 548.*

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List the symptoms of a barbiturate overdose.	Skeletal muscle tremors, diaphoresis, anxiety, tachycardia, orthostatic hypotension, and hyperreflexia. There is also a possibility of hyperthermia and complete cardiovascular collapse. The most deleterious problem is the occurence of grand mal seizures. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 548.*
What are the medically approved uses for amphetamines?	Amphetamines are used in the treatment of attention-deficit disorders and hyperactivity in children, and for the treatment of narcolepsy. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 549.*
What are the symptoms related to an overdose of amphetamines?	A psychotic state, anxiety, hypertension, cardiac dysrhythmias, tachycardia, diaphoresis, hyperthermia, hydriasis, and a decrease in gastrointestinal motility. Progressive irritability of the central nervous system occurs, which is characterized by hyperreflexia, hyperactivity, and on rare occasion, seizures. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 549.*
What percentage of patients with alcohol withdrawal syndrome suffer from delirium tremens?	Approximately 5% *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 544.*
Describe the treatment of an overdose related to oral amphetamine ingestion.	An overdose from oral amphetamines should be treated with gastric lavage or induced vomiting, followed by a cathartic and the administration of activated charcoal. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 549.*
Describe the withdrawal symptoms associated with the acute cessation of amphetamine use.	Increased appetite, weight gain, extreme lethargy, and suicidal depression. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 549.*
How should hypotension during anesthesia be treated in a patient with a history of chronic amphetamine abuse?	Hypotension should be treated with a direct-acting vasopressor such as epinephrine or phenylephrine. Responses to indirect-acting vasopressors may be less effective due to the catecholamine depletion caused by amphetamines. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 549.*
What are the symptoms related to salicylic acid overdose?	Nausea, vomiting, tinnitus, low blood sugar, fever, seizures, somnolence, coagulopathy, low CSF concentrations of glucose, direct respiratory center stimulation, liver dysfunction, and noncardiogenic pulmonary edema. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 552-553.*

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